Linux-Stack Based V2X Framework: SocketV2V
All You Need to Hack Connected Vehicles

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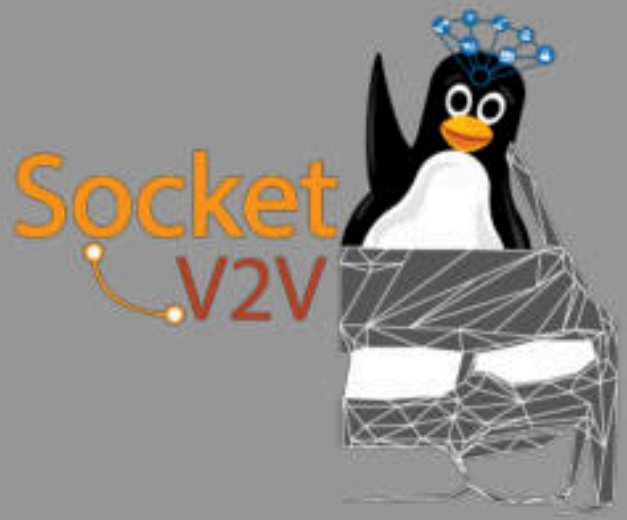


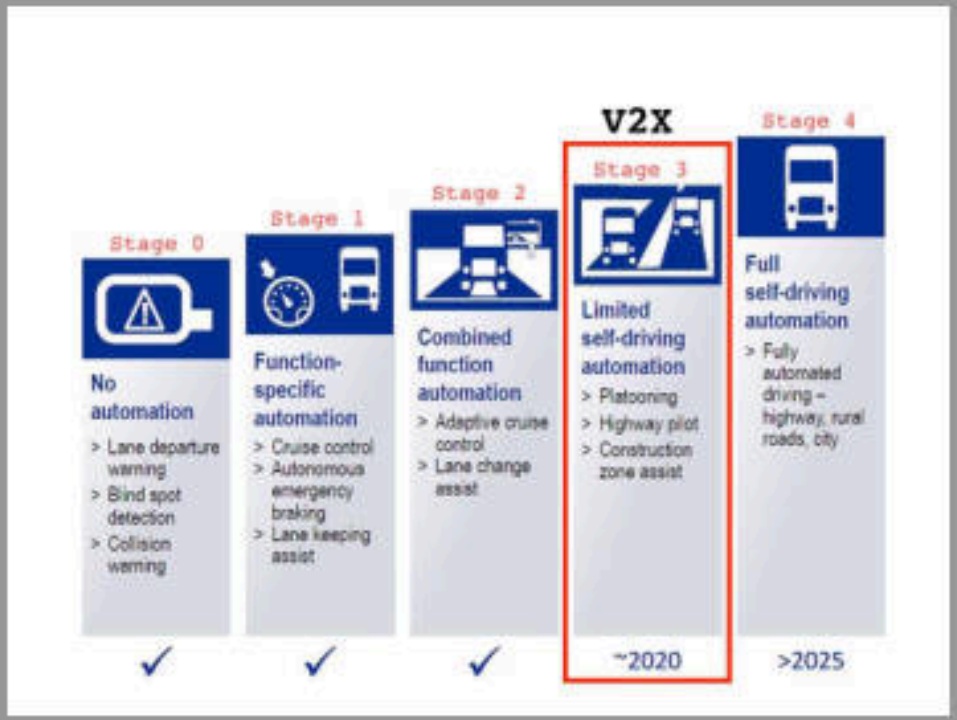
DEFCON 25: July 29, 2017

State of the World: (Semi)Autonomous Driving Technologies

- Vehicular automation widespread in global industry
- Automated driving technologies becoming accessible to general public

- Comms protocols used today in vehicular networks heavily flawed
- New automated technologies still using CANBUS and derivatives

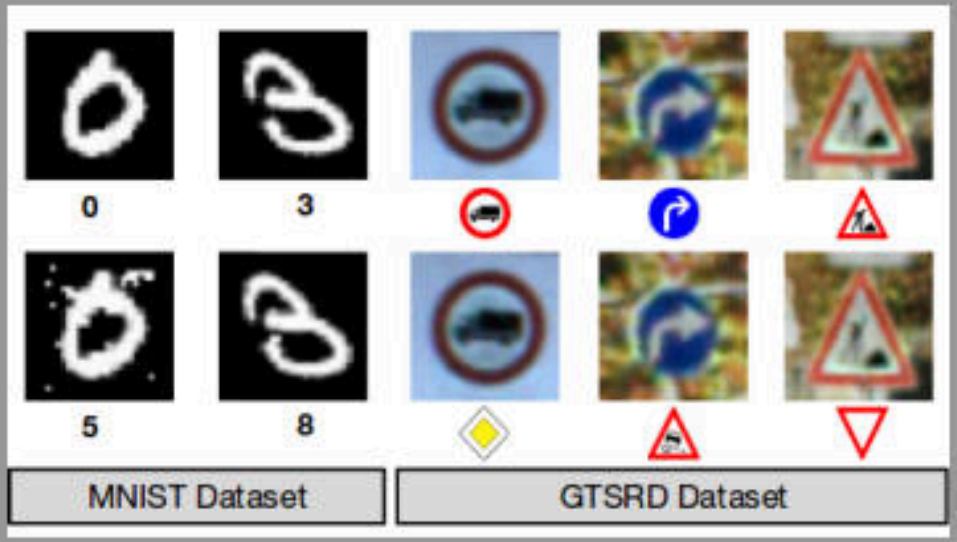
Stages of Autonomy



- Today: Stage 2 Autonomy - Combined Function Automation
- V2X: Stage 3 Autonomy - Limited Self-Driving Automation

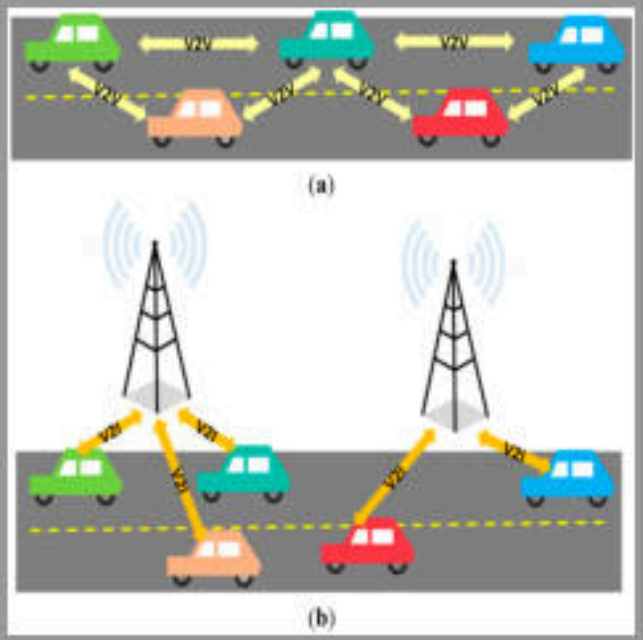
Barriers to Stage 3+ Autonomy

- Ownership of ethical responsibilities - reacting to safety-critical events
- Technological infrastructure
  - Installing roadside units, data centers, etc.
- Adaptive and intuitive machine-learning technology



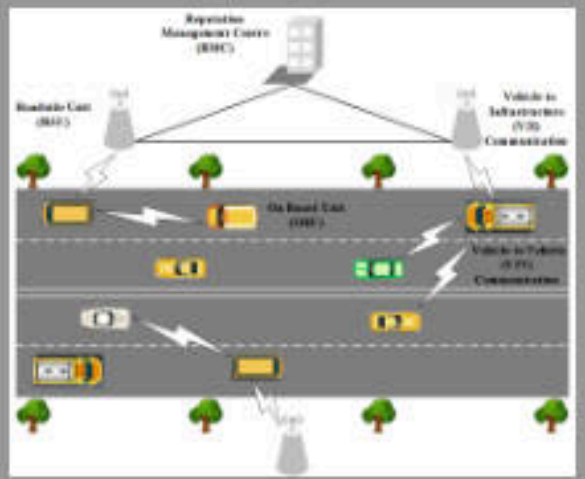
V2X Concept

- Vehicles and Infrastructure use WAVE over 5.8-5.9GHz adhoc mesh network to exchange state information
- Link WAVE/DSRC radios to vehicle BUS to enable automated hazard awareness and avoidance
- Technological bridge to fully autonomous vehicles



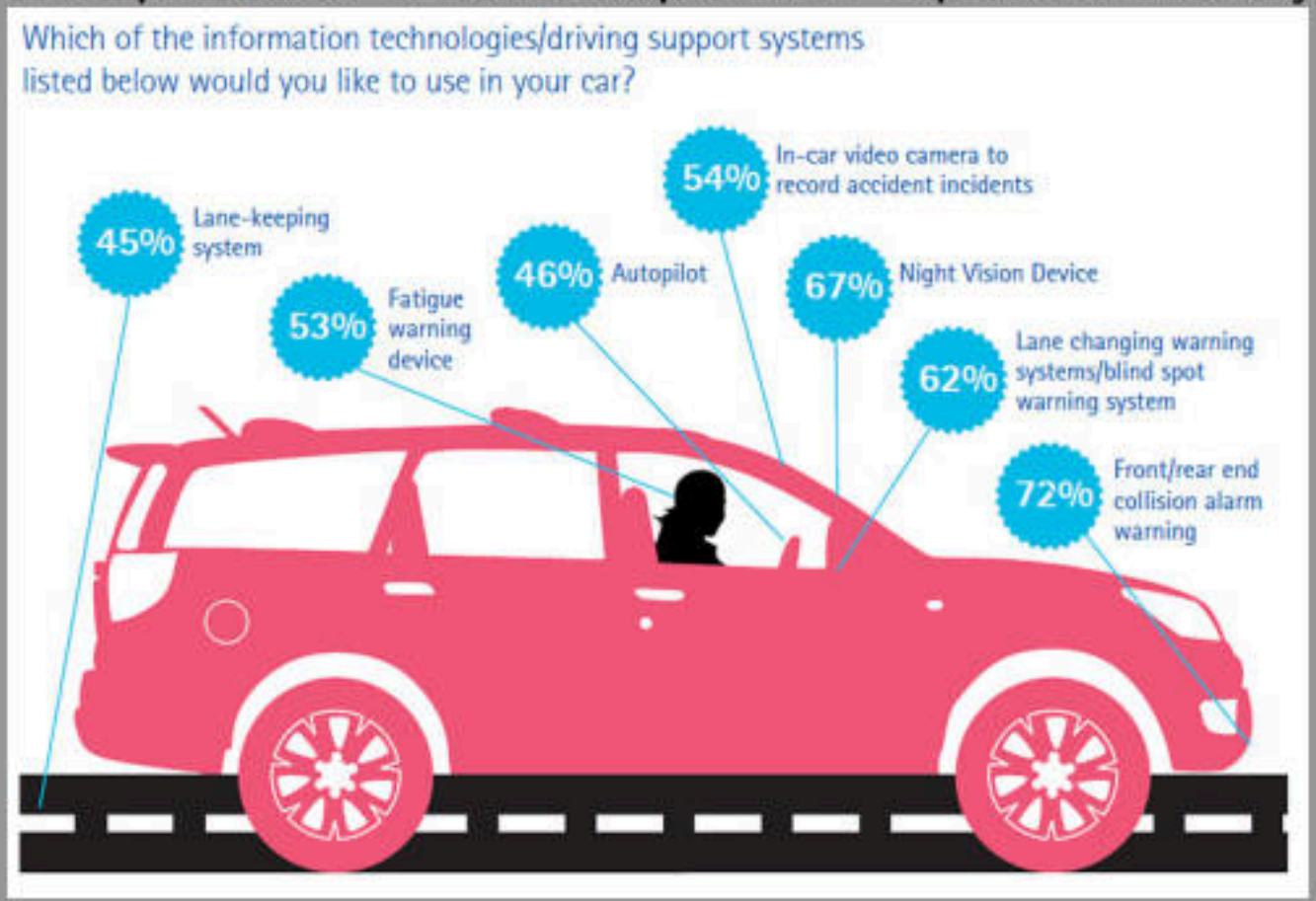
Critical Aspects of V2V

- High throughput vehicular ad hoc mesh network (VANET)
- Provide safety features beyond capability of onboard sensors
- Geared for homogeneous adoption in consumer automotive systems
- Easy integration with existing transportation infrastructure
- First application of stage 3 automation in consumer marketplace



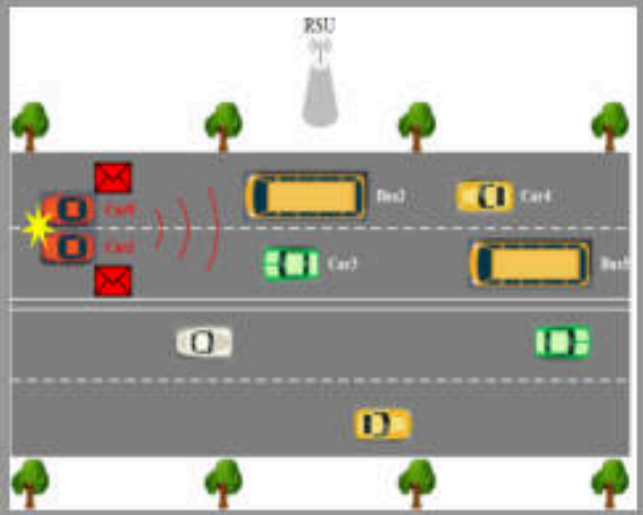
Impact of V2X Technologies

Transportation network impacts all aspects of society



Impact of V2X Technologies: on Consumers

- Safety benefits:
  - Prevent 25,000 to 592,000 crashes annually
  - Avoid 11,000 to 270,000 injuries
  - Prevent 31,000 to 728,000 property damaging crashes



- Traffic flow optimization:
  - 27% reduction for freight
  - 23% reduction for emergency vehicles
  - 42% reduction on freeway (with cooperative adaptive cruise control & speed harmonization)

Impact of V2X Technologies: Global Industry

- Scalable across industrial platforms
  - Optimize swarm functions
  - Improve exchange of sensor data
- Enhance/improve worker safety
  - Vehicle-to-pedestrian
  - Construction, agriculture, maintenance
- Improve logistical operations management
  - Think: air traffic control for trucks

Impact of V2X Technologies: Critical Infrastructure

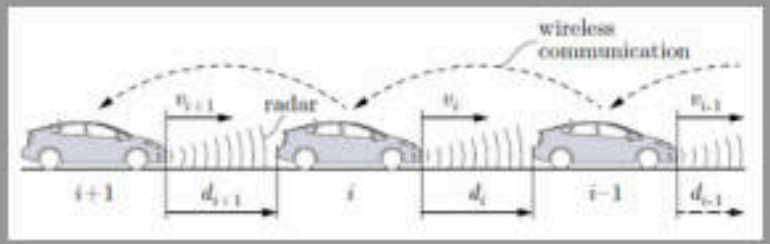
- Provide interface for infrastructure to leverage VANET as carrier
- Increase awareness of traffic patterns in specific regions
  - Analysis of network traffic facilitates improvements in civil engineering processes
- Fast widespread distribution of emergency alerts
- Reduce cost of public transit systems

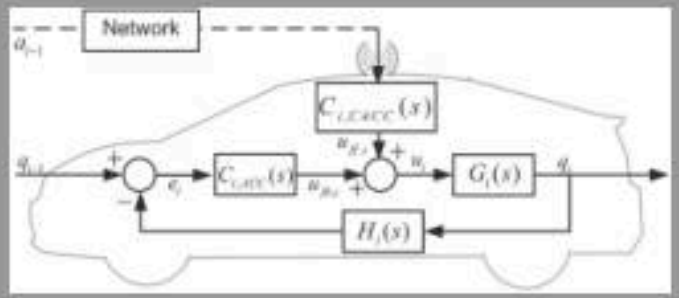
Impact of V2X Technologies: Automotive Security

- Wide open wireless attack vector into transportation network
- Injections easily propagate across entire VANET
- Wireless reverse engineering using 1609 and J2735
- Easy to massively distribute information (malware)

Technologies Using V2X

- Collision avoidance (Forward Collision Warning) systems
- Advanced Driver Assistance Systems (ADAS)

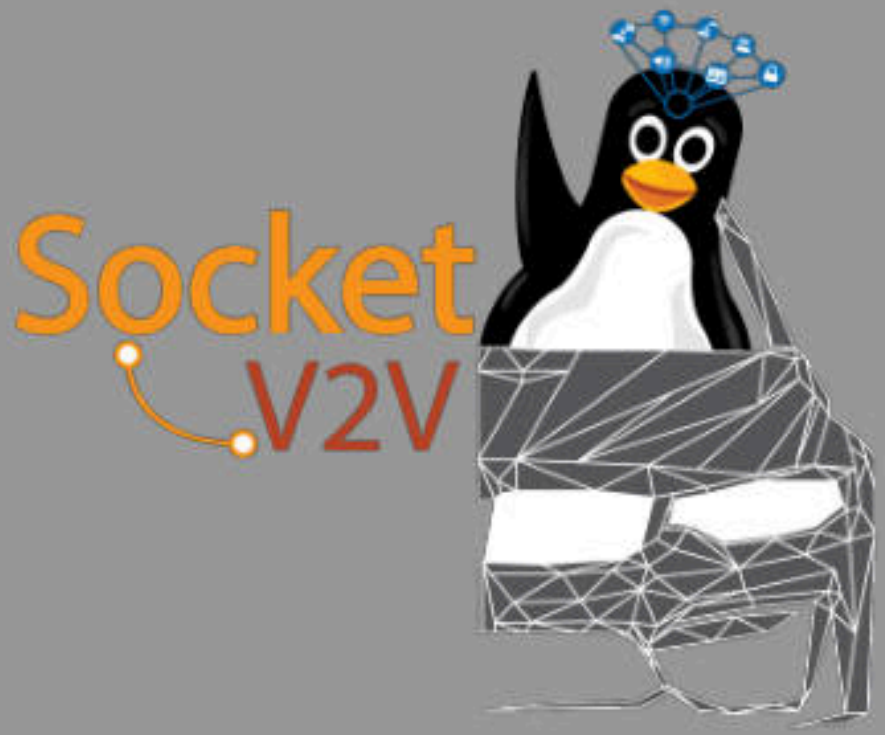




- Cooperative adaptive cruise control
- Automated ticketing and tolling

Vision of SocketV2V

- Security through obscurity leads to inevitable pwning
- Security community must be involved in development of public safety systems



- Catalyze development of secure functional connected systems
- Provide interface to VANET with standard COTS hardware

Background on SocketV2V

- Linux V2V development begins November 2015
- Large body of existing work found to be incomplete
  - No open-source implementation exists
  - Attempts at integration in linux-wireless since 2004
- Abandon efforts to patch previous attempts mid-2016
- Two years of kernel debugging later, V2V is **real**

Motivation for V2X Development

- Current standards for onboard vehicle communications not designed to handle VANET
- Increase in automation $\Rightarrow$ increase in attack surface
- Auto industry calling for proprietary solutions
  - Leads to a monopolization of technology
  - Standards still incomplete and bound for change, proprietary solutions become obsolete
  - Multiple alternative standards being developed independently across borders
- Imminent deployment requires immediate attention

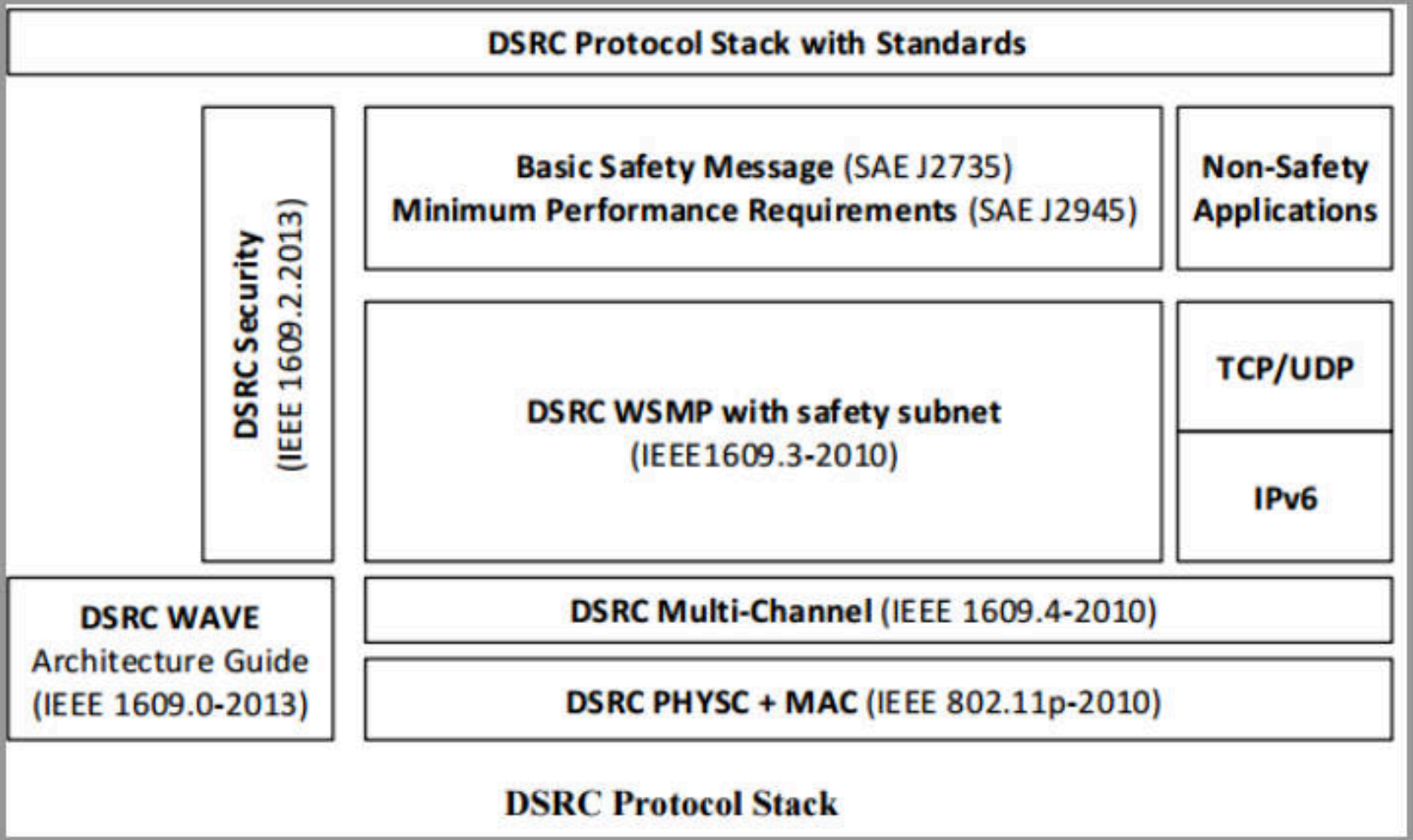
Lessons Learned from Previous Epic Failure

1. You keep calling yourself a kernel dev, I do not think it means what you think it means

```
/**
 * ieee80211_is_ocb - check if the frame is 802.11p compatible
 * @fc: frame control bytes in little-endian byteorder
 */
/* We accept:   MGMT: TSF, action
                *               CTL: ~{PSPOLL,CFEND,CFENDACK}
                *               DATA: data, null, QoS data, QoS null
                */
static inline bool ieee80211_is_ocb(__le16 fc) {
        if(ieee80211_is_data(fc)) {
                if(!ieee80211_is_data_present(fc) &&
                   !ieee80211_is_nullfunc(fc) &&
                   !ieee80211_is_data_qos(fc) &&
                   !ieee80211_is_qos_nullfunc(fc))
                        return false;
        } else if(ieee80211_is_mgmt(fc)) {
                if(!ieee80211_is_action(fc))
                        return false;
        } else if(ieee80211_is_ctl(fc)) {
                if(ieee80211_is_pspoll(fc) ||
                   ieee80211_is_cfend(fc) ||
                   ieee80211_is_cfendack(fc))
                        return false;
        } else {
                /* we don't support other frame types yet */
                //return false; /* WTF */
                return true;
        }
        return true;
}
```

2. Sharing is caring: closed-source development leads to failure
3. Standards committees need serious help addressing unprecedented levels of complexity in new and future systems

V2X Stack Overview



DSRC Protocol Stack

V2X Stack Overview: 802.11p

Wireless Access in Vehicular Environments

- Amendment to IEEE 802.11-2012 to support WAVE/DSRC
- PHY layer of V2X stack
- No association, no authentication, no encryption
- Multicast addressing with wildcard BSSID = {ff:ff:ff:ff:ff:ff}
- 5.8-5.9GHz OFDM with 5/10MHz subcarriers

Parameters	IEEE 802.11p
Channel bandwidth	10 MHz
Bit rate (Mbps)	3, 4.5, 6, 9, 12, 18, 24, 27
Modulation Mode	BPSK, QPSK, 16QAM, 64QAM
Number of subcarriers	52
Symbol duration	8 µs
Guard Interval Time	1.6 µs

IEEE 1609

WAVE Short Message Protocol (WSMP)

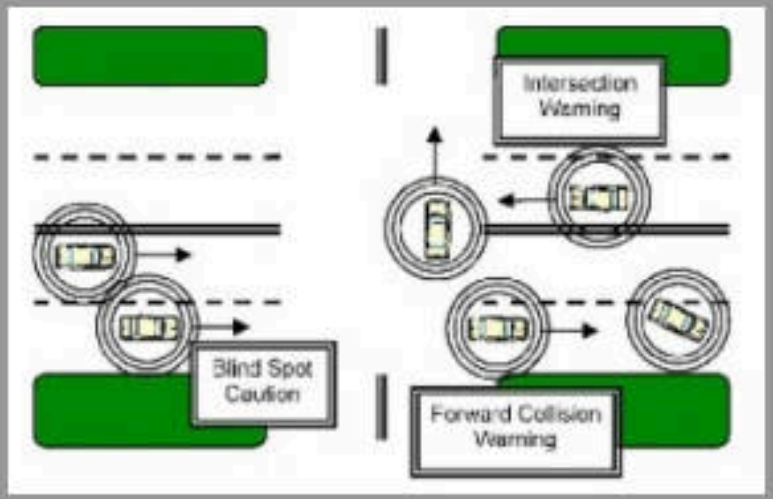
- 1609.2 Security Services
  - PKI, cert revocation, misbehavior reporting
- 1609.3 Networking Services
  - Advertisements, message fields
- 1609.4 Multi-Channel Operation
  - Channel sync, MLMEX
- 1609.12 Identifier Allocations
  - Provider service IDs

IEEE 1609: WAVE Short Message

Field Name				Length (octets)	Value (hex)	Description
WSMP-N-Header	*Subtype, Option Indicator, WSMP Version*			1	0B	Subtype = 0 (4 bits) Option Ind = 1 (1 bit) Version = 3 (3 bits)
	WAVE Information Element Extension	*Count*		1	03	Info Elem Count = 3
		Info Element 1	*Channel Number*	3	0F 01 AC	WAVE Element ID = 15 WAVE Elem Length = 1 Channel: 172
		Info Element 2	*Data Rate*	3	10 01 0C	WAVE Element ID = 16 WAVE Elem Length = 1 Data rate: 6 Mb/s
		Info Element 3	*Transmit Power Used*	3	04 01 9E	WAVE Element ID = 4 WAVE Elem Length = 1 30 dBm
	TPID			1	00	Address Info (PSID) only, no Info Elem Ext field present.
WSMP-T-Header	*Provider Service Identifier*			3	C0 03 05	PSID: 0pC0-03-05
	WSM Length			1	0D	Length = 13 Length of WSM Data
WSM Data	*WSM Data*			13	48 65 6C 6C 6F 20 57 6F 72 6C 64 21 00	ASCII content: 'Hello World!' 0x48 = H 0x65 = e 0X6C = l etc.

V2X Stack Overview: SAE J2735

- Message dictionary specifying message formats, data elements
  - Basic safety message, collision avoidance, emergency vehicle alert, etc.
  - ASN1 UPER specification
  - Also supports XML encoding



- Data element of Wave Short Message
- Application-layer component of V2X stack

State of V2X Standards: Evolution

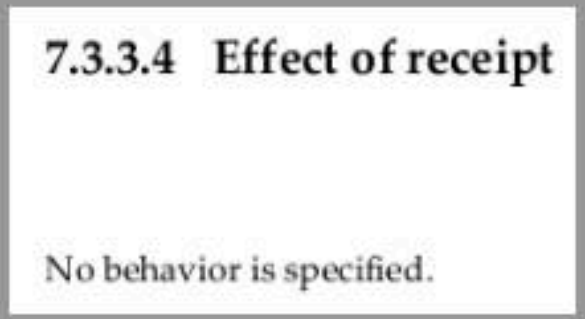
- WAVE drafted in 2005, J2735 in 2006
- WAVE revisions not backwards compatible
- IEEE 1609.2 incomplete
- J2735 revisions not backwards compatible
  - Encoding errors in J2735 ASN1 specification
- 3 active pilot studies by USDOT
- Experimental V2X deployment in EU
- Developmental status still in flux

Major Changes to the Standards

- Refactoring of security services to change certificate structure
- Refactoring of management plane to add services (P2PCD)
- Refactoring of application layer message encoding format
  - Multiple times: BER $\Rightarrow$ DER $\Rightarrow$ UPER
- Refactoring of application layer ASN1 configuration
- Revision of trust management system - still incomplete

(Possibly Unintentional) Obfuscation of the Standards

- No specification of handling for service management and RF optimization
- Minimal justification given for design choices



- Introduction of additional ambiguity in message parsing (WRA IEX block)
- Redlining of CRC data element in J2735 messages
- Refactoring of J2735 ASN1 to a non standard format
- Proposed channel sharing scheme with telecom

Subtleties in WAVE/J2735

- Ordering of certain fields not guaranteed in 1609
- Type incongruities in 1609
- Wave Information Element contains nested structures
- Channel synchronization mechanism based on proximal VANET traffic
- Channel switching necessary with one-antenna systems
- Implementation-specific vulnerabilities can effect entire network

V2X Attack Surfaces

- VANET accessible from a single endpoint
  - Attacks propagate easily across network
- Entry point to all systems connected to V2X infrastructure
  - Manipulation of traffic control systems: lights, bridges
  - Public transportation
  - Tolling and financial systems
- DSRC interface acts as entry point to onboard systems
  - Wireless access to vehicle control BUS
  - Transport malware across trafficked borders
- Privilege escalation in PKI
  - Hijacking emergency vehicle authority
  - Certificate revocation via RSU

Understanding the Adversary: Passive

- Determine trajectory of cars within some radius
  - Few stations required to monitor a typical highway
- Enumerate services provided by peers
- Characterize network traffic patterns within regions
- Uniquely fingerprint network participants independent of PKI
  - RF signature
  - Probe responses
  - Behavior patterns
- Perform arbitrage on economic markets

Understanding the Adversary: Active

- Denial of service, MITM
  - Impersonate infrastructure point
  - Manipulate misbehavior reports
- Disrupt vehicle traffic
  - Target specific platforms and individuals
  - Economic warfare: manipulation of supply/distribution networks
  - Behavior modeling and manipulation
- Privilege escalation in VANET
  - Parade as an emergency vehicle or moving toll station
    - Ad hoc PKI for application-layer services
  - Assume vehicle control via platooning service

V2X Threat Model: Applied

- Network traffic can be corrupted over-the-air
- Ad hoc PKI can allow certificate hijacking
- Diagnostic services in DSRC implementation expose vehicle control network
- Valid DSRC messages can pass malicious instructions to infotainment BUS
- Fingerprinting possible regardless of PKI pseudonym scheme
- Trust management services vulnerable to manipulation
  - Misbehavior reporting
  - Certificate revocation
  - Denial of P2P certificate distribution

Our Solution: Just Use Linux

- Platform-independent* V2X stack integrated in mainline Linux kernel
  - No proprietary DSRC hardware/software required - uses COTS hw
  - Extensible in generic Linux environment
  - *Currently supports ath9k
- Implements 802.11p, IEEE 1609.{3,4} in Linux networking subsystem
  - mac80211, cfg80211, nl80211
  - 1609 module to route WSMP frames
- Underlying 802.11 architecture compatible with Linux networking subsystem
  - Mainline kernel integration leads to immediate global deployment
  - This enables rapid driver integration

SocketV2V: Implementing 802.11p in Linux Kernel

- Add driver support for ITS 5.825-5.925GHz band
- Define ITS 5.8-5.9GHz channels

```
CHAN5G(5850, 38), /* Channel 170 */
/* ITA-G5B */
CHAN5G(5855, 39), /* Channel 171 */
CHAN5G(5860, 40), /* Channel 172 */
CHAN5G(5865, 41), /* Channel 173 */
CHAN5G(5870, 42), /* Channel 174 */
/* ITS-G5A */
CHAN5G(5875, 43), /* Channel 175 */
CHAN5G(5880, 44), /* Channel 176 */
CHAN5G(5885, 45), /* Channel 177 */
CHAN5G(5890, 46), /* Channel 178 */
CHAN5G(5895, 47), /* Channel 179 */
CHAN5G(5900, 48), /* Channel 180 */
CHAN5G(5905, 49), /* Channel 181 */
/* ITS-G5D */
CHAN5G(5910, 50), /* Channel 182 */
CHAN5G(5915, 51), /* Channel 183 */
CHAN5G(5920, 52), /* Channel 184 */
CHAN5G(5925, 53), /* Channel 185 */
```

SocketV2V: Implementing 802.11p in Linux Kernel

- Modify kernel and userspace local regulatory domain

```
country US: DFS-FCC
        (2402 - 2472 @ 40), (30)
        (5170 - 5250 @ 80), (17)
        (5250 - 5330 @ 80), (23), DFS
        (5735 - 5835 @ 80), (30)

        # For ITS-G5 evaluation
        (5850 - 5925 @ 20), (20), NO-CCK, OCB-ONLY
        (57240 - 63720 @ 2160), (40)
```

```
#define ATH9K_5GHZ_5150_5350    REG_RULE(5150-10, 5350+10, 80, 0, 30,\
                                        NL80211_RRF_NO_IR)
#define ATH9K_5GHZ_5470_5925    REG_RULE(5470-10, 5925+10, 80, 0, 30,\
                                        NL80211_RRF_NO_IR)
#define ATH9K_5GHZ_5725_5925    REG_RULE(5725-10, 5925+10, 80, 0, 30,\
                                        NL80211_RRF_NO_IR)
```

- Force use of user-specified regulatory domain

SocketV2V: Implementing 802.11p in Linux Kernel

- Enable filtering for 802.11p frames
- Require use of wildcard BSSID

```
case NL80211_IFTYPE_OCB:
        /* We accept:   MGMT: TSF, action
         *              CTL: ~{PSPOLL,CFEND,CFENDACK}
         *              DATA: data, null, QoS data, QoS null
         */
        if (!bssid)
                return false;
        if (!multicast &&
            !ether_addr_equal(sdata->vif.addr, hdr->addr3))     {
            printk("%s:%s Group addr have wildcard BSSID\n",__FILE__,__FUNCTION__);
                return false;
        }
        if (!multicast &&
            !ether_addr_equal(sdata->dev->dev_addr, hdr->addr1)) {
                printk("%s:%s bot mc and not addr1 addr \n",__FILE__,__FUNCTION__);
                return false;
        }
        if(!ieee80211_is_ocb(hdr->frame_control)) {
                printk("%s:%s wrong frame type for OCB\n",__FILE__,__FUNCTION__);
        }
        if (!rx->sta) {
                int rate_idx;
                if (status->encoding != RX_ENC_LEGACY)
                        rate_idx = 0;
                else
                        rate_idx = status->rate_idx;
                ieee80211_ocb_rx_no_sta(sdata, bssid, hdr->addr2,
                                        BIT(rate_idx));
        }
        return true;
```

SocketV2V: Userspace Utility Modifications for 802.11p

- Add **iw** command for joining 5GHz ITS channels using OCB

```
COMMAND(ocb, join, "<freq in MHz> <5MHz|10MHz>",
        NL80211_CMD_JOIN_OCB, 0, CIB_NETDEV, join_ocb,
        "Join the OCB mode network.");
```

```
static const struct chanmode chanmode[] = {
        { .name = "5MHz",
          .width = NL80211_CHAN_WIDTH_5,
          .freq1_diff = 0,
          .chantype = NL80211_CHAN_5MHZ },
        { .name = "10MHz",
          .width = NL80211_CHAN_WIDTH_10,
          .freq1_diff = 0,
          .chantype = NL80211_CHAN_10MHZ },
};
```

- Add **iw** definitions for 5/10MHz-width channels in OCB

- Use **iw** to join ITS spectrum with COTS WiFi hardware!

```
phy#1
        Interface wlxe091f53d73df
                ifindex 4
                wdev 0x100000001
                addr e0:91:f5:3d:73:df
                type outside context of a BSS
                channel 178 (5890 MHz), width: 10 MHz, center1: 5890 MHz
                txpower 17.00 dBm
```

SocketV2V: Implementing WAVE in Linux

Functions to pack, parse, and broadcast messages

- Relevant data structures
  - WSM, WSA, WRA, SII, CII, IEX
- Full control of fields
  - subtype, TPID, PSID, chan, tx power, data rate, location, etc.
  - Operating modes for setting degree of compliance to standard (strict, lax, loose)
- Channel switching, dispatch
- Netlink socket interface to userspace
- Userspace link to the 802.11p kernel routines
- Manage channel switching using iw

WAVE Usage

- Message fields set in struct wsmp_wsm
- WSMs encoded to bitstream through wsmp_wsm_encode
- WSM bitstream decoded through wsmp_wsm_decode
- Functions for generating random WSMP frames to specified compliance
- Opens socket to 802.11p wireless interface
- Inject WSMs with prefix EtherType 0x86DC

WAVE Structs: WSM

```
struct wsmp_wsm {
     /* N-Header */
     uint8_t subtype; /* 0 to 15 */
     uint8_t version; /* WSMP_VERSION */
     uint8_t tpid;    /* 0 to 5 */

     bool use_n_iex;
     struct wsmp_iex *n_iex;

     /* T-Header */
     union {
          uint32_t psid; /* 0x0 to 0xEFFFFFFF */

          struct {
               uint8_t src[2];
               uint8_t dst[2];
          } ports;
     };

     bool use_t_iex;
     struct wsmp_iex *t_iex;

     uint16_t len; /* 0 to 16383 */
     uint8_t *data;
};
```

- WAVE Short Message: message encapsulation and forwarding parameters (N-hop, flooding)

WAVE Structs: IEX

- Information Element Extension: optional fields for RF, routing, and services

```
struct wsmp_iex {
     uint16_t count; /* 1 to 255 */

     /* WSM Elements */
     uint8_t chan;       /* 0 to 200 */
     uint8_t data_rate; /* 2 to 127 */
     int8_t tx_pow;      /* -128 to 127 */

     /* SII Elements */
     struct wsmp_ie_psc psc;
     uint8_t ip[16];
     uint8_t port[2];
     uint8_t mac[6];
     int8_t rcpi_thres;        /* -110 to 0 */
     uint8_t count_thres;
     uint8_t count_thres_int; /* 1 to 255 */

     /* CII Elements */
     struct wsmp_ie_edca edca;
     uint8_t chan_access; /* 0 to 2 */

     /* WSA Elements */
     uint8_t repeat_rate;
     struct wsmp_ie_2d loc_2d;
     struct wsmp_ie_3d loc_3d;
     struct wsmp_ie_advert_id advert_id;

     /* WRA Elements */
     uint8_t sec_dns[16];
     uint8_t gateway_mac[6];

     bool use[WSMP_EID_CHANNEL_LOAD+1];

     /* Raw elements not defined in 1609.3-2016 */
     uint16_t raw_count;
     struct wsmp_ie_raw *raw;
};
```

(More) WAVE Structs

```
struct wsmp_wsa {
     uint8_t version;        /* 0 to 15 */
     uint8_t id;             /* 0 to 15 */
     uint8_t content_count; /* 0 to 15 */

     bool use_iex;
     struct wsmp_iex *iex;

     uint16_t sii_count; /* 0 to 31 */
     struct wsmp_sii **sis;

     uint16_t cii_count; /* 0 to 31 */
     struct wsmp_cii **cis;

     bool use_wra;
     struct wsmp_wra *wra;
};
```

```
struct wsmp_sii {
     uint32_t psid;
     uint8_t chan_index; /* 0 to 31 */

     bool use_iex;
     struct wsmp_iex *iex;
};

struct wsmp_cii {
     uint8_t op_class;  /* 802.11 */
     uint8_t chan;      /* 802.11 */
     int8_t tx_pow;     /* -128 to 127 */
     uint8_t adapt;     /* 0 to 1 */
     uint8_t data_rate; /* 0x02 to 0x7F */

     bool use_iex;
     struct wsmp_iex *iex;
};
```

```
struct wsmp_wra {
     uint8_t lifetime[2];   /* IETF RFC 4861 */
     uint8_t ip_prefix[16]; /* IETF RFC 4861 */
     uint8_t prefix_len;    /* IETF RFC 4861 */
     uint8_t gateway[16];
     uint8_t dns[16];

     bool use_iex;
     struct wsmp_iex *iex;
};
```

SocketV2V: Implementing J2735

```
/* BasicSafetyMessage */
struct BSM {
    enum DSRCmsgID msgID;
    struct {
        uint8_t msgCnt;
        uint32_t tid;
        uint16_t secMark;
        uint32_t lat;
        uint32_t longt;
        uint16_t elev;
        uint32_t accuracy;
        uint32_t txstate;
        uint16_t speed;
        uint16_t heading;
        uint8_t angle;
        uint64_t accelSet[4];
        uint16_t brakes;
        uint32_t size;
    } blob1;
};
```

```
struct BSM *bsm1;
bsm1 = calloc(sizeof(struct BSM),1);
bsm1->msgID = basicSafetyMessage;
bsm1->blob1.msgCnt = 0xAB;
bsm1->blob1.tid = 0x12345678;
bsm1->blob1.secMark = 0x6358;
bsm1->blob1.lat = 0x67BA2100;
bsm1->blob1.longt = 0x1A4B263D;
bsm1->blob1.elev = 0xAAAA;
bsm1->blob1.accuracy = 0x1221ABAB;
bsm1->blob1.txstate = 0x3;
bsm1->blob1.speed = 0xA2B;
bsm1->blob1.heading = 0x1212;
bsm1->blob1.angle = 0x42;
bsm1->blob1.accelSet[0] = 0x0A0B;
bsm1->blob1.accelSet[1] = 0x0C0D;
bsm1->blob1.accelSet[2] = 0x0FA;
bsm1->blob1.accelSet[3] = 0x1AFBC;
bsm1->blob1.brakes = 0xA134;
bsm1->blob1.size = 0x01323456;
```

SocketV2V: Implementing J2735

- Generate WSM
- Pack J2735 msg in WSM data element

```
struct wsmp_wsm *wsm1;
wsm1 = calloc(sizeof(struct wsmp_wsm),1);
wsm1->subtype = 0x0;
wsm1->version = 0x3;
wsm1->tpid = 0x00;
wsm1->use_n_iex = 0;
wsm1->psid = 0xC00305;
wsm1->use_t_iex = 0;
wsm1->len = sizeof(*bsm1);
wsm1->data = (uint8_t *) bsm1;
```

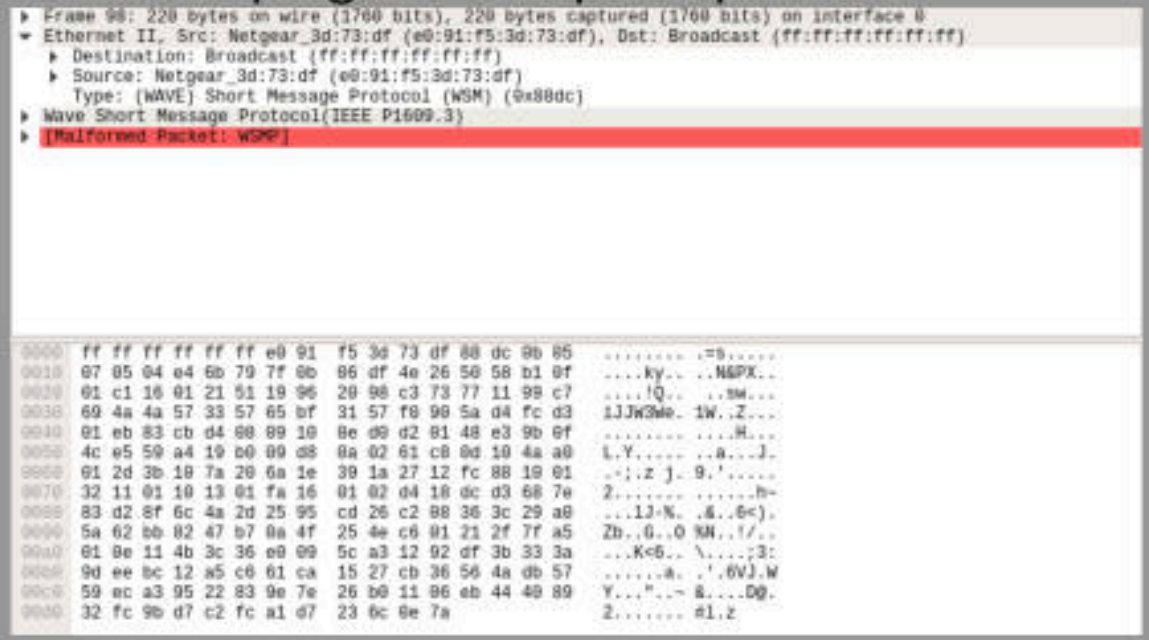
```
uint8_t *wsm = wsmp_wsm_encode(wsm1, &wsm_cnt, &wsm_err, WSMP_MODE);

if (wsm_err) {
     fprintf(stderr, "ERROR:\twsmp_if_send | wsm_err\n");
     return EXIT_FAILURE;
}

unsigned char frame[sizeof(struct ether_header) + wsm_cnt];
memcpy(frame, &hdr, sizeof(struct ether_header));
memcpy(frame + sizeof(struct ether_header), wsm, wsm_cnt);

if (pcap_inject(pcap, frame, sizeof(frame)) == -1) {
     pcap_perror(pcap, 0);
     return EXIT_FAILURE;
}
```

- Serialize WSM
- Tx using pcap_inject!

SocketV2V: Implementing J2735

```
98 2.885718542   Netgear_3d:73:df   Broadcast   WSMP   220 WAVE Short Message Protocol IEEE P1609.3[Malformed Pack..
```

Wireshark WSMP plugin incomplete per current 1609 encoding



SocketV2V: Implementing J2735

```
struct RoadSideAlert *rsal;
rsal = calloc(sizeof(struct RoadSideAlert),1);
rsal->msgID = roadSideAlert; // A 1 byte instance
rsal->msgCnt = 0xBA;
rsal->typeEvent = 0xDEFC;
rsal->position.lat = 0x67BA2100;
rsal->position.longt = 0x1A4B263D;
```

```
struct wsmp_wsm *wsm2;
wsm2 = calloc(sizeof(struct wsmp_wsm),1);
wsm2->subtype = 0x0;
wsm2->version = 0x3;
wsm2->tpid = 0x00;
wsm2->use_n_iex = 0;
wsm2->psid = 0xC00305;
wsm2->use_t_iex = 0;
wsm2->len = sizeof(*rsal);
wsm2->data = (uint8_t *) rsal;
```

```
struct EmergencyVehicleAlert *eval;
eval = calloc(sizeof(struct EmergencyVehicleAlert),1);
eval->msgID = emergencyVehicleAlert; // A 1 byte instance
eval->timeStamp = 0x515; // optional
eval->tid = 0x12345678;
eval->rsaMsg = rsal;
eval->basicType = 0x04;
```

```
struct wsmp_wsm *wsm3;
wsm3 = calloc(sizeof(struct wsmp_wsm),1);
wsm3->subtype = 0x0;
wsm3->version = 0x3;
wsm3->tpid = 0x00;
wsm3->use_n_iex = 0;
wsm3->psid = 0xC00305;
wsm3->use_t_iex = 0;
wsm3->len = sizeof(*eval);
wsm3->data = (uint8_t *) eval;
```

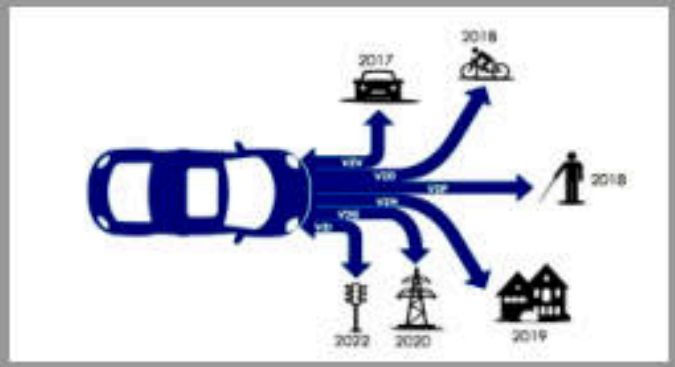
Future Pwning of ITS: You Wanna be a Master?

- Level 1: Denial of Service
  - Single-antenna DSRC systems susceptible to collision attack
- Level 2: DSRC spectrum sweep, enumerate proprietary (custom) services available per participant
- Level 3: Impersonate an emergency vehicle
- Level 4: Become mobile tollbooth
- Level 1337: Remotely execute platooning service
  - Assume direct control

Additional Forms of Pwning

Pandora's box:

- Mass dissemination of malware
- Passive surveillance with minimal effort
- Extract RF parameters for imaging
- Reverse engineer system architectures given enough data
- Epidemic propagation model
- Built in protocol switching
  - Exfiltration over comm bridges!



Developing Connected Vehicle Technologies

- Widespread access enables engagement of security (1337) community in standards development
- Interact with existing V2X infrastructure
  - Pressure manufacturers and OEMs to implement functional V2V
- Deploy ahead of market - experimental platforms
  - UAS, maritime, orbital, heavy vehicles
- Opportunity for empirical research: See what you can break
  - Straightforward to wardrive
  - Hook DIY radio (Pi Zero with 5GHz USB adapter) into CANBUS (for science ONLY)

Acknowledgments

References

- Check me out on GitHub: https://github.com/p3n3troot0r/socketV2V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